CJ Nour

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Professional Summary

Motivated Electrical Engineering new graduate from McMaster University with 2 years of professional experience at Tesla and Moneris. Passionate about controls engineering and automation, with a proven ability to deliver innovative solutions – ranging from industrial projects in autonomous systems to software-driven IoT solutions. Recognized for developing impactful, standardized designs and excelling in collaboration in cross-functional environments. Known for creativity, adaptability, and a results-driven mindset, I bring a rich blend of valuable practical experience, interpersonal abilities, and fresh academic knowledge to solve complex challenges and drive meaningful outcomes.

Core Competencies

- Control Systems
- Project Management
- Vendor Management
- PLC Programming
- Beckhoff TwinCAT3
- Object-Oriented Programming
- Python
- Microprocessors
- Computer Aided Design (CAD)

Professional Experience

Tesla

Manufacturing Controls Engineer Intern — Jan. 2024 – May 2024

- Spearheaded the development of an **RFID**-based monitoring system that aims to **reduce** tooling maintenance issues by **over 90%**. Integrated **PLC logic** and developed a custom **TwinCAT HMI dashboard** to track service history and collect tooling lifespan data on **multiple Gigafactory assembly lines**.
- Contributed to commissioning **3 automation** machines by leveraging **Beckhoff TwinCAT3 Structured Text** to program critical components, including **VFDs**, sensors, and robotic and pneumatic devices. Debugged field I/O networks using multiple industrial communication protocols (**TCP/IP, EtherCAT, Profinet**).
- Led multiple **R&D** initiatives to solve critical engineering challenges to enhance **Beckhoff PLC** and **FANUC robot** performance by experimenting with **computer vision systems**, **time-of-flight lasers**, **profilometers**. Authored **test result reports** and applied technical specifications to optimize device integration.

Material Planner Intern — Sep. 2023 – Dec. 2023

- Developed and sustained 3 **Excel VBA** tools to optimize inventory and logistics, **eliminating duplicate shipments by 100%**. Automated the parts reordering process, automated gap analyses for project needs, and streamlined shipping with auto-generated forms and emails - enhancing database operational efficiency.
- Leveraged these tools to **identify inefficiencies** in the existing third-party inventory system and built a business case to influence executives to create an internal software team to develop an improved, proprietary system.
- Facilitated international shipments, expedited missing parts, and streamlined kitting to **reduce technician downtime by 20%**, ensuring efficient workflows across Gigafactories and optimizing daily resource allocation.

Moneris

Technology Governance Intern — May 2023 – Aug. 2023

- Reduced the application database **by over 50%** through **cost center analysis** and a detailed **general ledger audit**. Retired outdated applications and underutilized software licenses, resulting in **significant cost savings**.
- Orchestrated collaboration across departments to gather, validate, and standardize data for **cloud migration planning**, ensuring a seamless transition phase and enhancing application scalability.
- Strengthened IT governance and compliance frameworks by aligning the application inventory with corporate standards to streamline software usage across teams, resulting in a **reduction of licensing costs by over 30%**.

Software Engineer Intern — Jun. 2022 – Apr. 2023

- Developed a tool **boosting productivity for 30+ employees** by enabling credit card transaction analytics. Built with **JavaScript** and **React**, the tool allows users to customize dashboards and manage widgets seamlessly.
- Improved code efficiency by **50%** by implementing **Redux** and a backend system hosted on **Parse**, optimizing personal and team tools by implementing reducers, utilizing dispatch, and **querying user database** properties.
- Collaborated in **agile** feedback sessions with the Scrum Master, QA team, Architecture team, and other stakeholders to identify and resolve platform issues related to our proprietary payment platform testing tool.

Professional Experience Continued

General Motors EcoCar EV Challenge

Propulsion Controls and Modeling Team Member Sep. 2024 – Present

- Developing **energy-efficient** and **customer-focused control features** as part of a competition sponsored by the **US Department of Energy** and **General Motors** involving 13 universities across North America.
- **Modelling control systems** in **MATLAB** and **Simulink** and deriving model requirements for a Cadillac Lyriq EV, ensuring precise alignment with vehicle specifications as outlined by GM and our team's performance goals.
- **Testing and validating control system requirements** using **Simulink**, ensuring compliance with benchmarks and functionality standards outlined by the competition organizers to achieve an **enhanced propulsion system**.

Projects

Autonomous Defense Drone System

- Developing an autonomous defense drone system to secure airspace by detecting, tracking, and intercepting unauthorized drones. The system integrates **multi-camera detection**, robotic arms designed in **CAD**, and an autonomous drone to provide a solution to drone-related threats.
- Utilizing **RF Tx/Rx** modules to enable communication between the camera hub system and the drones. Implementing **YOLOv8**, an **AI camera**, and **control algorithms** in **Python** to command the motors and autonomously track, pursue, and intercept unauthorized drones.

Autonomous RC Car

- Implemented self-driving algorithms coded in **C++ and Python** on an RC car to navigate through obstacles via **LiDAR** and **ultrasonic sensor** data processed through a **NVIDIA Jetson Nano** running on **Linux Ubuntu**.
- Coordinated **agile** team meetings as the Scrum Master to monitor progress, assign tasks, address challenges, and troubleshoot issues during development and testing phases.

Smart Curtain System

- Created a solution for sleep deprivation by designing an automated curtain. The app was coded in **React Native** and the backend is hosted on **AWS IoT** to store data for alarm functions and automated sunrise/sunset control.
- Implemented and deployed a **C++** program on an **Arduino Uno** to contact the backend server and control the stepper motor, enabling precise movement for rolling the curtain up and down automatically.

Wireless Dog Doorbell System

- Designed and developed an **affordable wireless doorbell system** with an **Arduino Uno**, enabling dogs to alert owners when they need to go outside, facilitating effective potty training through dog-to-human communication.
- Built a mobile app using **React Native** to send real-time push notifications to owners, ultimately enhancing communication and improving the effectiveness of the puppy training process.

Automated Vanity System

- Designed and fabricated a custom automated vanity system by 3D-modeling a recycled oil barrel in **Autodesk Inventor**. Cut and welded a panel from the barrel to a recycled chair and added wheels for smooth accessibility, and ensured the chair sat flush when retracted.
- Engineered a motorized mirror and a **Bluetooth** sound system by wiring a recycled car window regulator motor to a momentary rocker switch and a power supply with an **AC/DC rectifier**. Researched **torque-speed characteristics** and motor specifications to ensure the motor could support the load with the supplied power.

Education

B. Eng., Electrical Engineering Expected Apr. 2025

McMaster University Hamilton, ON

- **Relevant Coursework:** Control Systems, Intro to Mechatronics, Power Electronics, Electric Motor Drives, Electromagnetics I & II, Electric Devices & Circuits, Data Structures & Algorithms, Computer Architecture, High Performance Programming, Principles of Programming, Microprocessor Systems, Signals & Systems, Logic Design
- **Extracurriculars:** McMaster EcoCar EV Challenge PCM Team Member, GDSC 2023 Incubator Challenge
- **Special Interests:** Guitar, Music Production, 3D-Printing, Basketball, Skiing, Cycling, Running